Worker Bargaining Power and Market Tightness: Theory and Evidence

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What Determines Worker Bargaining Power?

Wages reflect the value an employment relationship creates.

When this value increases, only part of the gain accrues to wages.

A 10% rise in value added per worker is associated with a 0.5–1.5% rise in wages (Card et al., 2018).

The gains workers capture reflect their bargaining power.

Estimates suggest bargaining power varies:

- Across worker types.
- Across firms.
- Across countries.
- Over time.

But we lack an understanding of the mechanism.

What explains the variation in bargaining power?

Labour Market Tightness

A natural candidate is labour market tightness, $\theta = \dfrac{\text{Vacancies}}{\text{Jobseekers}}$.

The intuition is simple; formalising it is not.

Once a worker and a firm meet, why should other opportunities matter?

- **Nash bargaining:** fixed bargaining power; outside options affect wages. The bargaining process is not modelled explicitly.
- **Hall and Milgrom (2008):** the influence should be limited.
- **On-the-job search:** competing offers improve the worker's fallback position. Bargaining power remains fixed.
  Cahuc et al. (2006).

What This Paper Does: Continued Search

Workers and firms bargain by alternating offers.

They continue being paired with potential partners while negotiating.

With multiple pairings, they choose which partner to pursue.

Bargaining power comes from the ability to reject and counteroffer.

Tightness changes the value of rejecting: the partner may leave.

Tighter markets make it easier for workers to reject unattractive offers.

What This Paper Does: Empirical Evidence

In the model's wage equation:

Bargaining power = productivity pass-through into wages.

I use French matched employer–employee data.
I estimate the marginal product of labour at each firm.

I link wages to labour market tightness by occupation, area and year.

Productivity pass-through is higher in tighter labour markets.

One standard deviation higher tightness: 13% higher pass-through.

Roadmap

1. Theory: A model of wage bargaining
2. Empirical Evidence
3. Heterogeneity in Match Productivity
4. Conclusion

Theory

A Frictional Labour Market

There are N_u unemployed workers and N_v vacancies.

Labour market tightness is fixed and given by

$$\theta = \frac{N_v}{N_u}$$

$\lambda_w(\theta)$: probability a worker pairs with a firm. $\lambda_w'(\theta) > 0$

$\lambda_f(\theta)$: probability a firm pairs with a worker. $\lambda_f'(\theta) < 0$

Worker Bargaining Power

Once paired, a worker and a firm bargain over a wage w.

If they agree on a wage, they form a match and create a surplus.

Worker Surplus

$$S_w(w) = \underbrace{V_w(w)}_{\text{Employment}} - \underbrace{V_w^0}_{\text{Search}}$$

Firm Surplus

$$S_f(w) = \underbrace{V_f(w)}_{\text{Employment}} - \underbrace{V_f^0}_{\text{Search}}$$

Bargaining power β is the worker's share of total surplus.

$$\frac{\beta}{1-\beta} = \frac{S_w(w)}{S_f(w)} \qquad \text{in Nash bargaining}$$

The Bargaining Game

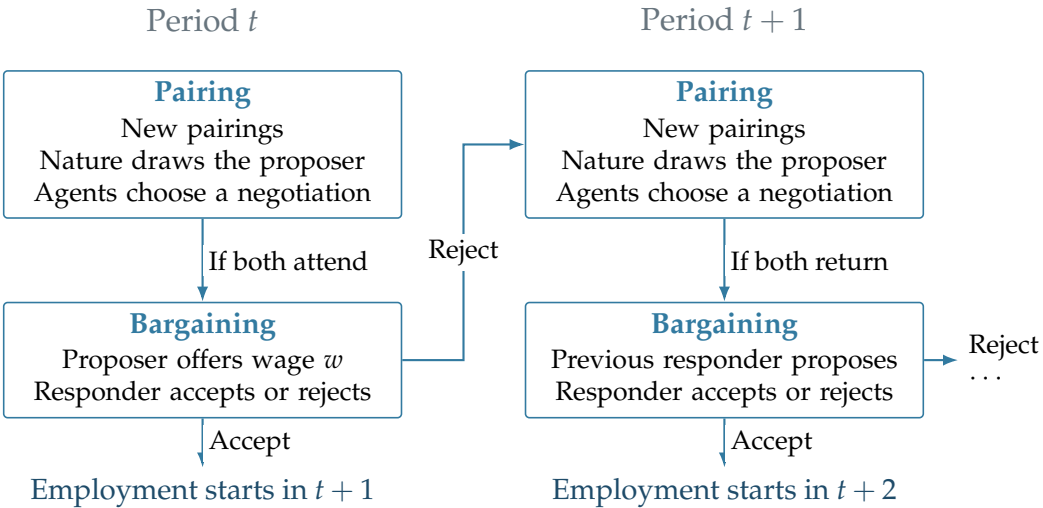
When paired, firms and workers bargain by **alternating offers**.

Nature draws the first proposer.

$\rightarrow$ In the baseline, each side proposes first with probability $1/2$.

- Agents know who would propose in each available negotiation.
- Each agent can attend at most one negotiation.
- If several are available, they choose which to attend.

Timing of a Negotiation



Markov Perfect Equilibrium

We solve for equilibrium in **bargaining** and **attendance** strategies.

Bargaining: wage offers w_f^*, w_w^* and an acceptance rule for each.

Attendance: a rule when multiple negotiations are available.

Bargaining

- Responders accept if accepting gives at least the value of rejecting.
- Proposers offer the wage that makes the responder indifferent.

Attendance

- Players prefer meetings in which they can propose.
- Conditional on this rule, they prefer new meetings.

The Worker's Indifference Condition

A worker receiving w_f^* is indifferent between accepting and rejecting.

$$V_w(w_f^*) = b + \underbrace{\delta}_{\text{Time passes}} \Bigg[\underbrace{\frac{\lambda_w}{2}}_{\substack{\text{New firm;} \\ \text{worker proposes}}} V_w(w_w^*) + \underbrace{\left(1 - \frac{\lambda_w}{2}\right)}_{\substack{\text{No new pairing} \\ \text{as proposer}}} \Big(\underbrace{(1-\lambda_f)}_{\substack{\text{Current firm} \\ \text{stays}}} V_w(w_w^*) + \underbrace{\lambda_f}_{\substack{\text{Current firm} \\ \text{leaves}}} V_w^0 \Big) \Bigg].$$

b is the unemployment payoff; $\delta \in (0,1)$ is the discount factor.

From Indifference to Surplus Shares

Subtract the value of continued search and rearrange.

$$S_w(w_f^*) = \delta \Bigg[S_w(w_w^*) - \underbrace{\lambda_f \left(1 - \frac{\lambda_w}{2}\right) S_w(w_w^*)}_{\substack{\text{Risk of not obtaining} \\ \text{the proposer surplus}}} - \underbrace{\frac{\lambda_w}{2}\left(S_w(w_w^*) + S_w(w_f^*)\right)}_{\text{Expected gains from search}} \Bigg].$$

The firm's surplus condition has the same two components.

Two Opposing Effects of Market Tightness

Lower Risk of Losing the Proposer Surplus $\uparrow \beta$

The firm is less likely to find a replacement worker.
The worker is more likely to find another firm as proposer.
The probability of securing w_w^* is higher.

Higher Value of Search Relative to Negotiation $\downarrow \beta$

Surplus is relative to search: $S_w(w) = V_w(w) - V_w^0$.
An unemployed worker benefits from any new pairing.
A worker already negotiating has less to gain from new pairings.
Tightness raises the relative value of search through this channel.

Bargaining Power and Market Tightness

Let $\bar{\beta}(\theta)$ be the worker's expected surplus share.

Each side proposes first with probability $1/2$.

$$\bar{\beta}(\theta) = \frac{1}{2}\left[1 + \frac{\delta(\lambda_w - \lambda_f)}{2 + \delta\left[1 + (1-\lambda_w)(1-\lambda_f)\right]}\right].$$

$$\boxed{\frac{\partial \bar{\beta}(\theta)}{\partial \theta} > 0}$$

Worker bargaining power rises with tightness.

The first channel dominates.

Equilibrium Wages

Let p be the output produced by a worker–firm match.

The expected equilibrium wage $\bar{w}$ is

$$\bar{w} = \bar{\beta}(\theta)p + (1-\delta)\Big[(1-\bar{\beta}(\theta))V_w^0 - \bar{\beta}(\theta)V_f^0\Big].$$

Holding search values fixed,

$$\frac{\partial \bar{w}}{\partial p} = \bar{\beta}(\theta) \qquad \text{and} \qquad \frac{\partial^2 \bar{w}}{\partial p\,\partial \theta} > 0.$$

Productivity pass-through rises with tightness.

Empirical Evidence

Data: Wages and Productivity

French matched employer–employee data, 2011–2023.

Workers and Wages

BTS–Postes

- Wages and hours.
- Occupation.
- Age and gender.
- Firm identifier.

Firm Productivity

FARE

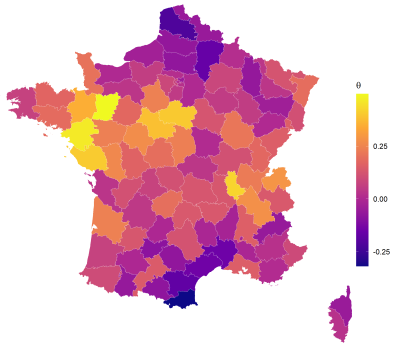
- Value added.
- Capital.
- Intermediate inputs.

I estimate hourly marginal product of labour (MPL) using Levinsohn–Petrin.

Each worker is linked to the productivity of their firm.

Data: Labour Market Tightness

DARES, the statistical office of the French Ministry of Labour.



- Occupation $\times$ location $\times$ year.
- 2011–2023.
- Locations are départements.
- Standardised over 2011–2022.
- The indicator combines
  - Vacancies relative to jobseekers.
  - Job-finding rates.
  - Recruitment difficulties.

Estimating Equation: Worker Level

I estimate hourly wages w as

$$w_{ijolt} = \alpha_1 \text{MPL}_{jt} + \alpha_2 \theta_{olt} + \alpha_3 (\text{MPL}_{jt} \times \theta_{olt}) + X'_{it}\gamma + \Phi_{jolt} + \varepsilon_{ijolt}.$$

- MPL_{jt}: firm-level hourly marginal product of labour.
- θ_{olt}: occupation $\times$ location $\times$ year tightness.
- X_{it}: age, age squared and gender.
- Φ_{jolt}: firm, occupation, location and year fixed effects.

Controls, year and location FEs $\longrightarrow$ control for search values.

$\alpha_3 > 0$**: productivity pass-through rises with tightness.**

Empirical Strategy: Instrumental Variables

Challenges

1) Tightness is an equilibrium outcome.
2) Firm productivity may be endogenous and is measured with error.

Labour Market Tightness

Lagged same-occupation tightness in other locations.

$$\theta^{\text{LOO}}_{ol,t-1} \;\rightarrow\; \theta_{olt}$$

Employment-weighted.

Leave-one-out design:
Azar et al. (2022); Börschlein et al. (2024).

Firm Productivity

Lagged marginal product of labour at the same firm.

$$\text{MPL}_{j,t-1} \;\rightarrow\; \text{MPL}_{jt}$$

Rent-sharing literature:
Card et al. (2018); Bell et al. (2022).

The product of the two IVs instruments the interaction.

Tightness and Productivity Pass-Through

	No Firm FE	Firm FE
Hourly MPL	0.0749***	0.0329***
	(0.0042)	(0.0052)
Tightness	0.0130	0.0112
	(0.0702)	(0.0435)
MPL × Tightness	0.0129***	0.0042***
	(0.0027)	(0.0012)
Observations	39,282,927	39,282,927
First-stage Wald F-statistics		
MPL	295.89	140.82
Tightness	2,150.30	2,480.70
MPL × Tightness	754.14	763.01

2SLS; hourly wages. Worker controls and occupation, location and year fixed effects. SEs clustered by firm and local labour market. $^{***}p < 0.01$.

Labour Market Tightness at Hiring

Workers hired after 2011: we know tightness at the start of employment.

	No Firm FE		Firm FE	
Tightness measured	Current	At Hiring	Current	At Hiring
Hourly MPL	0.0787***	0.0783***	0.0368***	0.0367***
	(0.0052)	(0.0060)	(0.0075)	(0.0077)
Tightness	0.0493	0.2792***	−0.0163	0.3082***
	(0.0793)	(0.0630)	(0.0562)	(0.0308)
MPL × Tightness	0.0069*	0.0148***	−0.0003	0.0030*
	(0.0035)	(0.0037)	(0.0015)	(0.0015)
Observations	6,516,075	6,516,075	6,489,935	6,489,935

All columns include worker controls and occupation, location and year fixed effects.
SEs clustered by firm and local labour market. $^{*}p < 0.05$; $^{***}p < 0.001$.

Heterogeneity in Match Productivity

Firm Heterogeneity

The baseline model studies bargaining power **across labour markets**.

As long as match surplus is the same, the mechanism is unchanged by:

- Multiple workers.
- Other production inputs.
- Decreasing returns to labour.

When match surplus differs across firms, bargaining power can also vary **within labour markets**.

Bargaining with Heterogeneous Firms

Let firms differ in productivity p.

Workers now consider the productivity p' of a new partner.

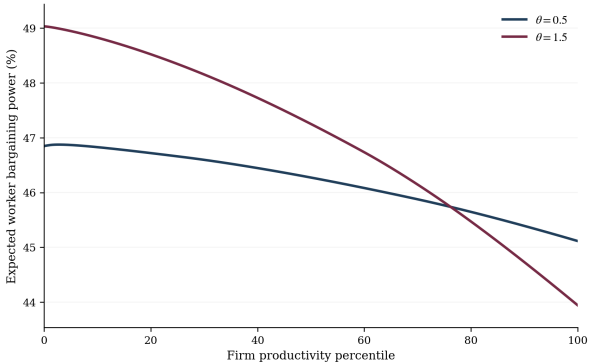
Workers' equilibrium strategies are characterised by productivity cutoffs:

$$\text{Switch from } p \text{ to } p' \quad \Longleftrightarrow \quad p' \geq p^*(p, \text{proposer draws}).$$

For given proposer draws, let $w(p)$ be the equilibrium wage.

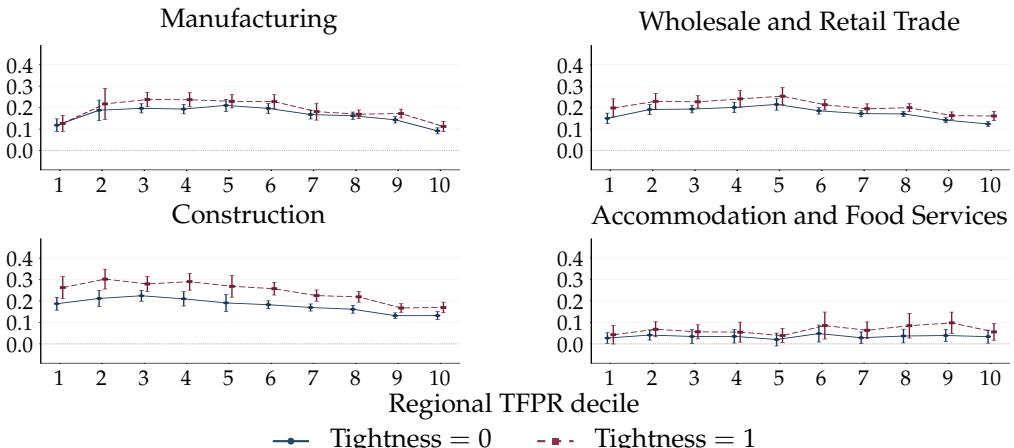
- Wages increase with firm productivity: $\dfrac{dw(p)}{dp} > 0$.
- Hence productivity cutoffs also increase: $\dfrac{dp^*(p)}{dp} > 0$.

Firm Productivity and Bargaining Power



- Bargaining power is lower at the top of the productivity distribution.
- The effect of tightness falls with productivity and can turn negative.

Productivity Pass-Through Across Firms: Empirical Evidence



Conclusion

Conclusion

Worker bargaining power is an **equilibrium outcome**.

Workers and firms continue to search while negotiating.

In tighter markets, firms are less likely to leave after a rejection.

Rejecting an unattractive offer becomes a more credible threat.

Tighter markets increase productivity pass-through into wages.

From average tightness to one standard deviation above:

3.3 $\longrightarrow$ 3.7 cents per euro of hourly productivity (+13%).

Conclusion

These findings have implications for **policy evaluation**.

Bargaining power shapes productivity pass-through into wages.

It also shapes wage responses to hiring subsidies and unemployment benefits.

→ These policies change the value of search.

Policy effects can therefore differ across markets with different tightness.

In general equilibrium, policies can affect both search values and tightness.

Both channels need to be considered for a complete policy evaluation.